

Levocet syrup 0.5mg/ml

COMPOSITION

Active Ingredient:

Levocetizirine Dihydrochloride 0.5mg/ml

Excipients:

Sodium citrate	11mg/ml
Sucralose	0.35mg/ml
D-Sorbitol Solution (70%)	350mg/ml
Grape Flavor	1mg/ml
Citric Acid	3.5mg/ml
Sodium Benzoate	1mg/ml
Purified water	Added to 1ml

DOSAGE FORM

Oral Solution

PRODUCT DESCRIPTION

Levocet syrup 0.5mg/ml is a colorless to pale yellow clear solution. A sweet taste of sugar, with grape flavor and a smell of grape note.

CLINICAL PHARMACOLOGY

Pharmacodynamics

Pharmacotherapeutic group: Antihistamines for systemic use, piperazine derivatives.

Mechanism of action

Levocetizirine, the (R) enantiomer of cetirizine, is a potent and selective antagonist of peripheral H1-receptors.

Binding studies revealed that levocetizirine has high affinity for human H1-receptors ($K_i = 3.2 \text{ nmol/l}$). Levocetizirine has an affinity 2-fold higher than that of cetirizine ($K_i = 6.3 \text{ nmol/l}$).

Levocetizirine dissociates from H1-receptors with a half-life of $115 \pm 38 \text{ min}$.

After single administration, levocetizirine shows a receptor occupancy of 90% at 4 hours and 57% at 24 hours.

Pharmacokinetics

Absorption

Levocetizirine is rapidly and extensively absorbed following oral administration. In adults, peak plasma concentrations are achieved 0.9 h after dosing. Steady state is achieved after two days.

Peak concentrations are typically 270 ng/ml and 308 ng/ml following a single and a repeated 5 mg o.d. dose, respectively. The extent of absorption is dose-independent and is not altered by food, but the peak concentration is reduced and delayed.

Distribution

No tissue distribution data are available in humans, neither concerning the passage of levocetizirine through the blood-brain-barrier. In rats and dogs, the highest tissue levels are found in liver and kidneys, the lowest in the CNS compartment.

In humans, levocetizirine is 90% bound to plasma proteins. The distribution of levocetizirine is restrictive, as the volume of distribution is 0.4 l/kg.

Biotransformation

The extent of metabolism of levocetizirine in humans is less than 14% of the dose and therefore differences resulting from genetic polymorphism or concomitant intake of enzyme inhibitors are expected to be negligible. Metabolic pathways include aromatic oxidation, N- and O- dealkylation and taurine conjugation. Dealkylation pathways are primarily mediated by CYP 3A4 while aromatic oxidation involved multiple and/or unidentified CYP isoforms. Levocetizirine had no effect on the activities of CYP isoenzymes 1A2, 2C9, 2C19, 2D6, 2E1 and 3A4 at concentrations well above peak concentrations achieved following a 5 mg oral dose.

Due to its low metabolism and absence of metabolic inhibition potential, the interaction of levocetizirine with other substances, or vice-versa, is unlikely.

Elimination

The plasma half-life in adults is $7.9 \pm 1.9 \text{ hours}$. The half-life is shorter in small children.

The mean apparent total body clearance in adults is 0.63 ml/min/kg . The major route of excretion of levocetizirine and metabolites is via urine, accounting for a mean of 85.4% of the dose.

Excretion via feces accounts for only 12.9% of the dose. Levocetizirine is excreted both by glomerular filtration and active tubular secretion.

Special population

Renal impairment

The apparent body clearance of levocetizirine is correlated to the creatinine clearance. It is therefore recommended to adjust the dosing intervals of levocetizirine, based on creatinine clearance in patients with moderate and severe renal impairment. The amount of levocetizirine removed during a standard 4-hour hemodialysis procedure was $< 10\%$.

Pediatric population

The paediatric patient with oral administration of a single dose of 5mg Levocetizirine in 14 children age 6 to 11 years with body weight ranging between 20 and 40kg show that Cmax and AUC values are about 2-fold greater in paediatric population than in healthy adult. The mean Cmax was 450 ng/ml, occurring at a mean time of 1.2 hours, weight-normalized, total body clearance was 30% greater, and the elimination half-life 24% shorter in this paediatric population than in adults.

Elderly

The total body clearance on elderly was approximately 33% lower compared to that in younger adults. The disposition of racemic cetirizine is dependent on renal function rather than on age.

This was also applicable to levocetizirine and cetirizine which are both predominantly excreted in urine. Therefore, the levocetizirine dose should be adjusted in accordance with renal function in elderly patients.

Gender

The half-life was slightly shorter in women than in men, however the body weight-adjusted oral clearance in women appears to be comparable to that in men. The same daily doses and dosing intervals are applicable for men and women with normal renal function.

Race

As levocetizirine is primarily renal excreted, and there are no important racial differences in creatinine clearance, pharmacokinetic characteristics of levocetizirine are not expected to be different across races. No race-related differences in the kinetics of racemic cetirizine have been observed.

Hepatic impairment

Patients with chronic liver diseases (hepatocellular, cholestatic, and biliary cirrhosis) given 10 or 20 mg of the racemic compound cetirizine as a single dose had a 50% increase in half-life along with a 40% decrease in clearance compared to healthy person.

Pharmacokinetic / pharmacodynamic relationship

The action on histamine-induced skin reactions is out of phase with the plasma concentrations.

INDICATION

Levocetizirine is indicated for the symptomatic treatment of allergic rhinitis (including persistent allergic rhinitis) and chronic idiopathic urticaria.

RECOMMENDED DOSE

Instruction for use

Appropriate volume of oral solution should be measured with the oral syringe, and poured in a spoon or in a glass of water. The oral solution must be taken orally immediately after dilution, and may be taken with or without food.

Adults and adolescents 12 years and above

The daily recommended dose is 5 mg (10 ml of solution).

Elderly

Adjustment of the dose is recommended in elderly patients with moderate to severe renal impairment (see Patients with renal impairment below).

Children aged 2 to 6 years:

The daily recommended dose is 2.5mg to be administered in 2 intakes of 1.25mg (2.5 ml of solution twice daily).

Children aged 6 to 12 years

The daily recommended dose is 5 mg (10 ml of solution). Due to the lack of data in this population, the administration of the product to infants and toddlers aged less than 2 years is not recommended.

Patients with renal impairment:

The dosing intervals must be individualized according to renal function. Refer to the following table and adjust the dose as indicated. To use this dosing table, an estimate of the patients creatinine clearance (CLcr) in ml/min is needed. The CLcr (ml/min) may be estimated from serum creatinine (mg/dl) determination using the following formula: $CL_{cr} = \frac{\{140 - \text{age (years)}\} \times \text{weight (kg)}}{72 \times \text{serum creatinine (md/dl)}}$ x {0.85 for women}

Dosing adjustments for patients with impaired renal function:

Group	Creatinine clearance (ml/min)	Dosage and frequency
Normal	≥ 80	5 mg once daily
Mild	50 – 79	5 mg once daily
Moderate	30 – 49	5 mg once every 2 days
Severe	< 30	5 mg once every 3 days
End-stage renal disease - Patients undergoing dialysis	< 10	Contra-indicated

Patients with hepatic impairment

No dose adjustment is needed in patients with solely hepatic impairment. In patients with hepatic impairment and renal impairment, adjustment of the dose is recommended (see patients with renal impairment above).

Duration of use

Intermittent allergic rhinitis (symptoms < 4 days/week or during less than 4 weeks) has to be treated according to the disease and its history; it can be stopped once the symptoms have disappeared and can be restarted again when symptoms reappear. In case of persistent allergic rhinitis (symptoms > 4 days/week and during more than 4 weeks), continuous therapy can be proposed to the patient during the period of exposure to allergens.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATION

- Hypersensitivity to the active substance, to cetirizine, to hydroxyzine, to any other piperazine derivatives or to any of the other excipients.
- Severe renal impairment at less than 10ml/min creatinine clearance.

WARNING & PRECAUTIONS

- Precaution is recommended with concurrent intake of alcohol.
- Caution should be taken in patients with epilepsy and patients at risk of convulsion as levocetizirine may cause seizure aggravation.
- Caution should be taken in patients with predisposing factors of urinary retention (e.g. spinal cord lesion, prostatic hyperplasia) as levocetizirine may increase the risk of urinary retention.
- Response to allergy skin tests are inhibited by antihistamines and a wash-out period (of 3 days) is required before performing them.
- Pruritus may occur when levocetizirine is stopped even if those symptoms were not present before treatment initiation. The symptoms may resolve spontaneously. In some cases, the symptoms may be intense and may require treatment to be restarted. The symptoms should resolve when the treatment is restarted.

INTERACTIONS WITH OTHER MEDICAMENTS

A small decrease in the clearance of cetirizine (16%) was observed with theophylline (400 mg once a day); while the disposition of theophylline was not altered by concomitant cetirizine administration.

The extent of exposure to cetirizine was increased by about 40% while the disposition of ritonavir was slightly altered (-11%) further to concomitant cetirizine administration.

The extent of absorption of levocetizirine is not reduced with food, although the rate of absorption is decreased.

In sensitive patients, the concurrent administration of cetirizine or levocetizirine and alcohol or other CNS depressants may cause additional reductions in alertness and impairment of performance.

PREGNANCY & LACTATION

Pregnancy

The use of levocetizirine may be considered during pregnancy, if necessary.

Breast-feeding

Cetirizine, the race mate of levocetizirine, has been shown to be excreted in human. Therefore, the excretion of levocetizirine in human milk is likely. Adverse reactions associated with levocetizirine may be observed in breastfed infants. Therefore, caution should be exercised when prescribing levocetizirine to lactating women.

SIDE EFFECTS

Adverse reactions from post-marketing experience are per System Organ Class and per frequency.

The frequency is defined as follows: very common; common; uncommon; rare; very rare, not known.

- Immune system disorders:
 - Not known: hypersensitivity including anaphylaxis
- Metabolism and nutrition disorders:
 - Not known: increased appetite
- Psychiatric disorders:

Not known: aggression, agitation, hallucination, depression, insomnia, suicidal ideation, nightmare

- Nervous system disorders:

Not known: convulsion, paraesthesia, dizziness, syncope, tremor, dysgeusia

- Ear and labyrinth disorders:

Not known: vertigo

- Eyes disorders:

Not known: visual disturbances, blurred vision, oculogyration

- Cardiac disorders:

Not known: palpitations, tachycardia

- Respiratory, thoracic and mediastinal disorders:

Not known: dyspnea

- Gastrointestinal disorders:

Not known: nausea, vomiting, diarrhea

- Hepatobiliary disorders:

Not known: hepatitis

- Renal and urinary disorders:

Not known: dysuria, urinary retention

- Skin and subcutaneous tissue disorders:

Not known: angioneurotic edema, fixed drug eruption, pruritus, rash, urticaria

- Musculoskeletal, connective tissues, and bone disorders:

Not known: myalgia, arthralgia

- General disorders and administration site conditions:

Not known: edema

- Investigations:

Not known: weight increased, abnormal liver function tests

SYMPTOMS & TREATMENT OF OVERDOSE

Symptoms

Symptoms of overdose may include drowsiness in adults. In children, agitation and restlessness may initially occur, followed by drowsiness.

Management of overdoses

There is no known specific antidote to levocetizirine.

Should overdose occur, symptomatic or supportive treatment is recommended. Gastric lavage may be considered shortly after ingestion of the drug. Levocetizirine is not effectively removed by hemodialysis.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE

Some patients could experience somnolence, fatigue and asthenia under therapy with levocetizirine. Therefore, patients intending to drive, engage in potentially hazardous activities or operate machinery should take their response to the medicinal product into account.

INSTRUCTION FOR USE

Appropriate volume of oral solution should be measured with the oral syringe, and poured in a spoon or in a glass of water. The oral solution must be taken orally immediately after dilution, and may be taken with or without food.

STORAGE CONDITONS

Store below 30°C.

The in-use storage condition : Discard 2 weeks after opening

PACKAGING

Levocet syrup 0.5mg/ml available in 60ml pack size, packed in brown PET plastic bottle (made up from plastic material). 1 bottle of 60ml Levocet Syrup 0.5mg/ml was packed in one printed box made up from cardboard material.

SHELF LIFE

3 Years

Manufactured by:

Chen Ho Pharmaceutical Co., Ltd. Sinying Plant

No.23, Singong Road, Jiafang Vill., Sinying Dist., Tainan City, 730 Taiwan, R.O.C.

Product Registration Holder:

Asumed Biotech Sdn Bhd

SS-21-08, Stellar Suites, Jalan Puteri 4/7, Bandar Puteri Puchong, 47100 Puchong, Selangor, Malaysia. Tel: +6017-806 4882

Distributed by:

AlPharma Healthcare Sdn Bhd (1304105-X)

a member of the **pharm-d** health science group

No. 83, Jalan I-Park 1/9, Kawasan Perindustrian I-Park, Bandar Indahpura, 81000 Kulai, Johor, Malaysia. Tel: +6017-806 4886

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